

Intangible Assets

Intangible Assets				
Intangible Asset Issues	Types of Intangibles	Impairment of Intangibles	Research and Development Costs	Presentation of Intangibles and Related Items
• Characteristics • Valuation • Amortization	• Marketing-related • Customer-related • Artistic-related • Contract-related • Technology-related • Goodwill	• Limited-life intangibles • Reversal of impairment loss • Indefinite-life intangibles other than goodwill • Goodwill	• Identifying R&D • Accounting for R&D • Similar costs	• Intangible assets • R&D costs

INTANGIBLE ASSET ISSUES

Characteristics

1. Identifiable.
2. Lack physical existence.
3. Not monetary assets.

Coca-Cola Company's (USA) success comes from its secret formula for making Coca-Cola, not its plant facilities.

Normally classified as non-current asset.

Common types of intangibles:

- | | |
|------------------------|---------------------------|
| ◆ Patent | ◆ Trademark or trade name |
| ◆ Copyright | ◆ Customer list |
| ◆ Franchise or license | ◆ Goodwill |

INTANGIBLE ASSET ISSUES

Valuation

Purchased Intangibles

- ◆ Recorded at cost.
- ◆ Includes all acquisition costs plus expenditures to make the intangible asset ready for its intended use.
- ◆ Typical costs include:
 - ▶ Purchase price.
 - ▶ Legal fees.
 - ▶ Other incidental expenses.

INTANGIBLE ASSET ISSUES

Valuation

Internally Created Intangibles

- ◆ Companies expense all research phase costs and some development phase costs.
- ◆ Certain development costs are capitalized once economic viability criteria are met.
- ◆ **IFRS** identifies several specific criteria that must be met before development costs are capitalized.

Underlying Concepts

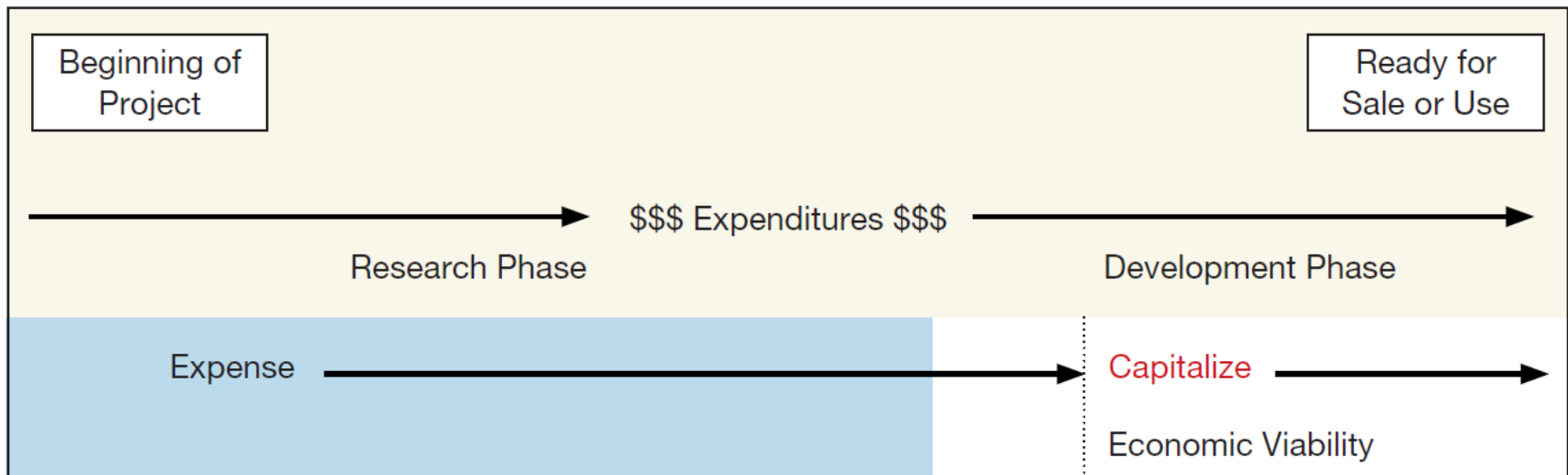


The controversy surrounding the accounting for R&D expenditures reflects a debate about whether such expenditures meet the definition of an asset. If so, then an “expense all R&D costs” policy results in overstated expenses and understated assets.

INTANGIBLE ASSET ISSUES

Internally Created Intangibles

ILLUSTRATION 12-1
Research and
Development Stages



INTANGIBLE ASSET ISSUES



Underlying Concepts

The terms “carrying amount,” “carrying value,” and “book value” are all used to refer to the cost less accumulated amortization of intangible assets.

Amortization of Intangibles

Limited-Life Intangibles

- ◆ Amortize by systematic charge to expense over useful life.
- ◆ Credit asset account or accumulated amortization.
- ◆ Useful life should reflect the periods over which the asset will contribute to cash flows.
- ◆ Amortization should be cost less residual value.
- ◆ Companies must evaluate the limited-life intangibles annually for impairment.

INTANGIBLE ASSET ISSUES

Amortization of Intangibles

Indefinite-Life Intangibles

- ◆ No foreseeable limit on time the asset is expected to provide cash flows.
- ◆ Must test indefinite-life intangibles for impairment at least annually.
- ◆ No amortization.

INTANGIBLE ASSET ISSUES

Amortization of Intangibles

ILLUSTRATION 12-2
Accounting Treatment
for Intangibles

Type of Intangible	Manner Acquired		Amortization	Impairment Test
	Purchased	Internally Created		
Limited-life intangibles	Capitalize	Expense*	Over useful life	Compare recoverable amount to carrying value.
Indefinite-life intangibles	Capitalize	Expense*	Do not amortize	Compare recoverable amount to carrying value.
	* (Except for certain limited costs that meet recognition criteria.)			

TYPES OF INTANGIBLE ASSETS

Six Major Categories:

- | | |
|------------------------|-------------------------|
| (1) Marketing-related. | (4) Contract-related. |
| (2) Customer-related. | (4) Technology-related. |
| (3) Artistic-related. | (4) Goodwill. |

TYPES OF INTANGIBLE ASSETS

Marketing-Related Intangible Assets

◆ Examples:

- ▶ **Trademarks** or **trade names**, newspaper mastheads, Internet domain names, and non-competition agreements.

- ◆ In Ethiopia trademark or trade name has legal protection for indefinite number of 7 year renewal periods (Trademark Registration and Protection Proclamation No.501/2006)..

- ◆ Capitalize purchase price.
- ◆ No amortization.



TYPES OF INTANGIBLE ASSETS

Customer-Related Intangible Assets

◆ Examples:

▶ Customer lists, order or production backlogs, and both contractual and non-contractual customer relationships.

◆ Capitalize acquisition costs.

◆ Amortized to expense over useful life.

TYPES OF INTANGIBLE ASSETS

Illustration: Green Market Inc. acquires the customer list of a large newspaper for €6,000,000 on January 1, 2015. Green Market expects to benefit from the information **evenly** over a three-year period. Record the purchase of the customer list and the amortization of the customer list at the end of each year.

Jan. 1 2015	Customer List	6,000,000
	Cash	6,000,000
Dec. 31 2015	Amortization Expense	2,000,000
2016	Customer List *	2,000,000
2017		

TYPES OF INTANGIBLE ASSETS

Artistic-Related Intangible Assets

◆ Examples:

- ▶ Plays, literary works, musical works, pictures, photographs, and video and audiovisual material.

◆ **Copyright** granted for the life of the creator plus 70 years.

◆ In Ethiopia copyright is granted for the life of the creator plus 50 years (**Copyright and Neighboring Rights Protection Proclamation No. 410/2004**).

◆ Capitalize costs of acquiring and defending.

The
WALT DISNEY
Company

and

Mickey
Mouse

◆ Amortized to expense over useful life if less than the legal life.

TYPES OF INTANGIBLE ASSETS

Contract-Related Intangible Assets



◆ Examples:

- ▶ Franchise and licensing agreements, construction permits, broadcast rights, and service or supply contracts.

- ◆ **Franchise** (or license) with a limited life should be amortized as operating expense over the life of the franchise.
- ◆ Franchise with an indefinite life should be carried at cost and not amortized.



TYPES OF INTANGIBLE ASSETS

Technology-Related Intangible Assets

- ◆ **Examples:**



- ▶ Patented technology and trade secrets granted by a government body.

- ◆ **Patent** gives holder exclusive use for a period of 20 years.

- ◆ In Ethiopia patent gives the holder exclusive use for a period of 15 years (**Inventions, Minor Inventions and Industrial Designs Proclamation No. 123/1995**).

- ◆ Capitalize costs of purchasing a patent.

- ◆ Expense any R&D costs in developing a patent.

- ◆ Amortize over legal life or useful life, whichever is shorter.

Canon



TYPES OF INTANGIBLE ASSETS

Illustration: Harcott Co. incurs \$180,000 in legal costs on January 1, 2015, to successfully defend a patent. The patent's useful life is 20 years, amortized on a straight-line basis. Harcott records the legal fees and the amortization at the end of 2015 as follows.

Jan. 1 Patents 180,000

 Cash 180,000

Dec. 31 Patent Amortization Expense 9,000

 Patents (or Accumulated Amortization)
 9,000

Patent Amortization Expense = $(\$180,000 \div 20) = \$9,000$

TYPES OF INTANGIBLE ASSETS

Goodwill

Conceptually, represents the future economic benefits arising from the other assets acquired in a business combination that are not individually identified and separately recognized.

Only recorded when an entire business is purchased.

Goodwill is measured as the excess of ...

*cost of the purchase **over** the fair value of the identifiable net assets (assets less liabilities) purchased.*

Internally created goodwill should not be capitalized.
--

RECORDING GOODWILL

Illustration: Feng, Inc. decides that it needs a parts division to supplement its existing tractor distributorship. The president of Feng is interested in buying Tractorling Company. The illustration presents the statement of financial position of Tractorling Company.

ILLUSTRATION 12-4

TRACTORLING CO. STATEMENT OF FINANCIAL POSITION AS OF DECEMBER 31, 2015			
Assets		Equities	
Property, plant, and equipment, net	\$153,000	Share capital	\$100,000
Inventories	42,000	Retained earnings	100,000
Receivables	35,000	Current liabilities	55,000
Cash	25,000		
Total assets	<u>\$255,000</u>	Total equities	<u>\$255,000</u>

RECORDING GOODWILL

Illustration: Feng investigates Tractorling's underlying assets to determine their fair values.

ILLUSTRATION 12-5

Fair Values	
Property, plant, and equipment, net	\$205,000
Patents	18,000
Inventories	122,000
Receivables	35,000
Cash	25,000
Liabilities	(55,000)
Fair value of net assets	<u><u>\$350,000</u></u>

Tractorling Company decides to accept Feng's offer of \$400,000. What is the value of the goodwill, if any?

RECORDING GOODWILL

Illustration: Determination of Goodwill.

ILLUSTRATION 12-6

Assigned to purchase price of \$400,000	→ Property, plant, and equipment, net	\$ 205,000
	→ Patents	18,000
	→ Inventories	122,000
	→ Receivables	35,000
	→ Cash	25,000
	→ Liabilities	<u>(55,000)</u>
	Fair value of net identifiable assets	\$ 350,000
	Purchase price	<u>(400,000)</u>
	Value assigned to goodwill	<u><u>\$ 50,000</u></u>

Book Value = \$200,000

Fair Value = \$350,000

Purchase Price = \$400,000

Revaluation
\$150,000

Goodwill
\$50,000

RECORDING GOODWILL

Illustration: Feng records this transaction as follows.

Property, Plant, and Equipment	205,000
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Patents	18,000
---------	--------

Inventory	122,000
-----------	---------

Accounts Receivables	35,000
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Cash	25,000
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Goodwill	50,000
----------	--------

Liabilities	55,000
-------------	--------

Cash	400,000
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RECORDING GOODWILL

Goodwill Write-Off

- ◆ Goodwill considered to have an **indefinite life**.
- ◆ Should **not** be amortized.
- ◆ Only adjust carrying value when goodwill is impaired.

Bargain Purchase

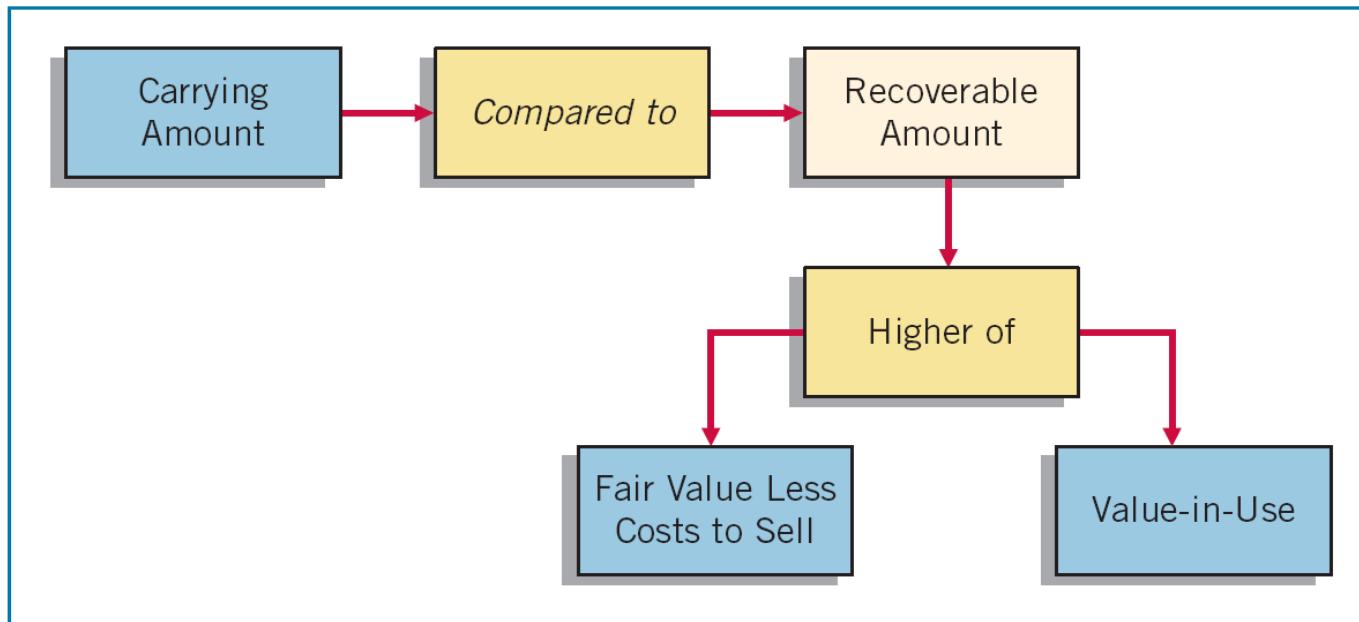
- ◆ Purchase price **less than** the fair value of net assets acquired.
- ◆ Amount is recorded as a **gain** by the purchaser.

IMPAIRMENT OF INTANGIBLE ASSETS

Impairment of Limited-Life Intangibles

A long-lived tangible and intangible asset are **impaired** when a company is not able to recover the asset's carrying amount either through using it or by selling it.

The impairment loss is the carrying amount of the asset less the recoverable amount of the impaired asset.



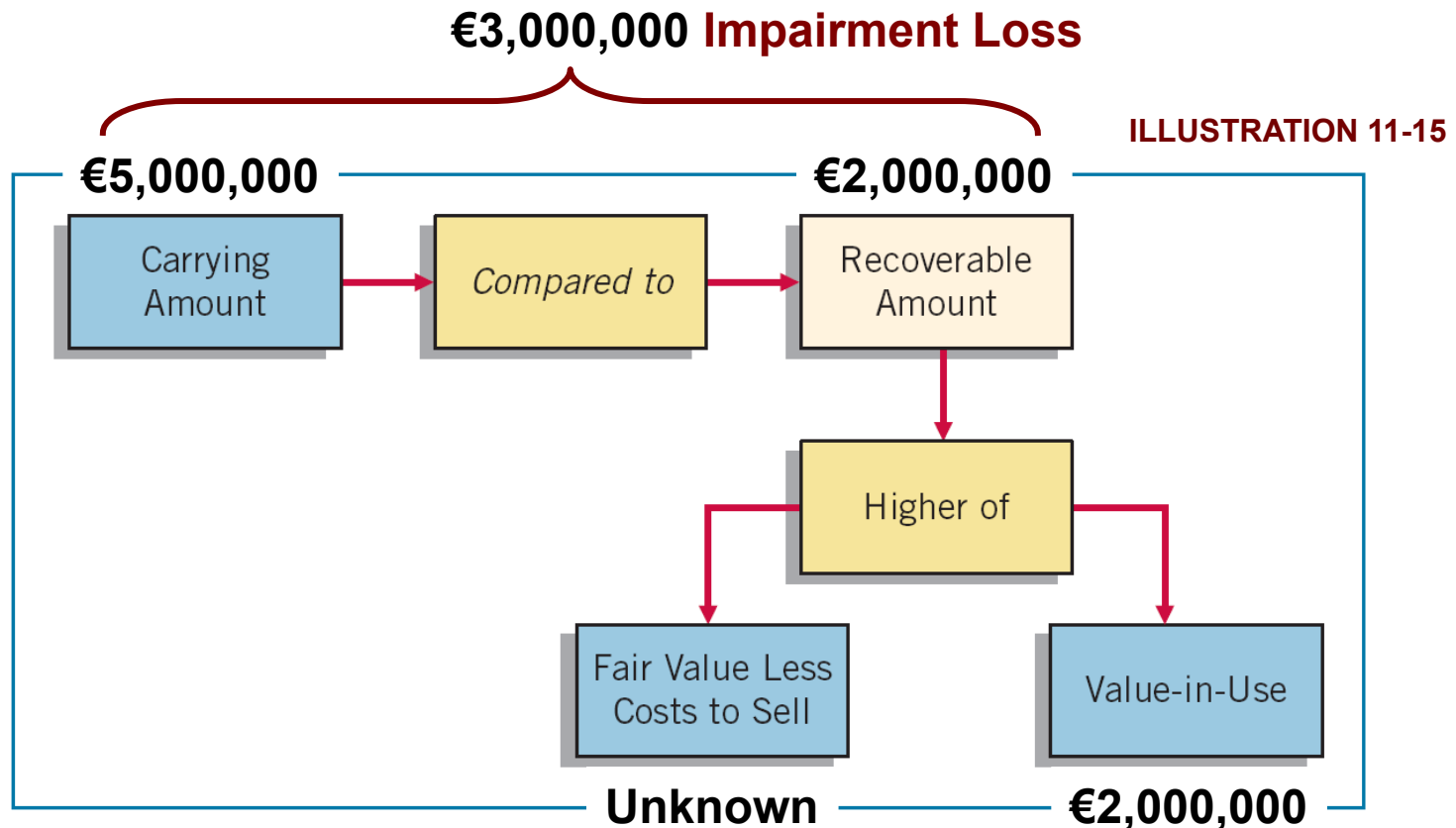
IMPAIRMENT OF INTANGIBLE ASSETS

Illustration: Lerch, Inc. has a patent on how to extract oil from shale rock, with a carrying value of €5,000,000 at the end of 2014. Unfortunately, several recent non-shale-oil discoveries adversely affected the demand for shale-oil technology, indicating that the patent is impaired. Lerch determines the recoverable amount for the patent, based on value-in-use (because there is no active market for the patent). Lerch estimates the patent's value-in-use at €2,000,000, based on the discounted expected net future cash flows at its market rate of interest.

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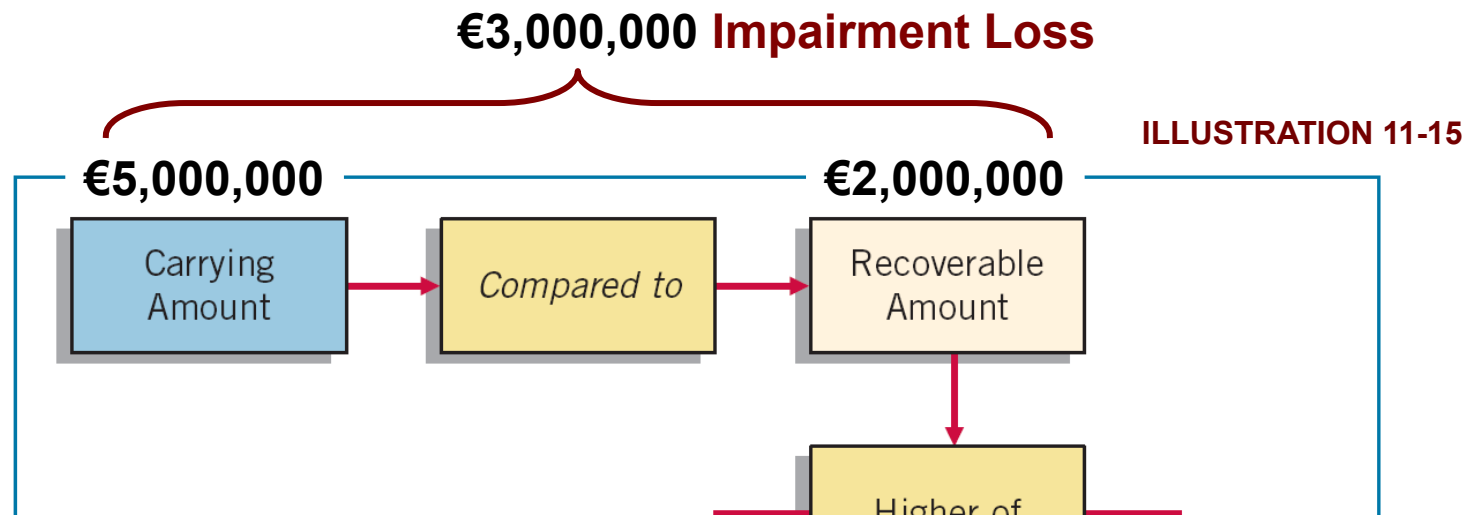
IMPAIRMENT OF INTANGIBLE ASSETS

Calculate the impairment loss (based on value-in-use).



IMPAIRMENT OF INTANGIBLE ASSETS

Calculate the impairment loss (based on value-in-use).



Lerch makes the following entry to record the impairment.

Loss on Impairment	3,000,000
Patents	3,000,000

IMPAIRMENT OF INTANGIBLE ASSETS

Reversal of Impairment Loss

Illustration: The carrying value of the patent after impairment is €2,000,000. Lerch's amortization is €400,000 ($€2,000,000 \div 5$) over the remaining five years of the patent's life. The amortization expense and related carrying amount after the impairment is shown below:

ILLUSTRATION 12-8

<u>Year</u>	<u>Amortization Expense</u>	<u>Carrying Amount</u>	
2015	€400,000	€1,600,000	(€2,000,000 – €400,000)
2016	400,000	1,200,000	(€1,600,000 – €400,000)
2017	400,000	800,000	(€1,200,000 – €400,000)
2018	400,000	400,000	(€800,000 – €400,000)
2019	400,000	0	(€400,000 – €400,000)

IMPAIRMENT OF INTANGIBLE ASSETS

Reversal of Impairment Loss

Early in 2016, based on improving conditions in the market for shale-oil technology, Lerch remeasures the recoverable amount of the patent to be €1,750,000. In this case, Lerch reverses a portion of the recognized impairment loss.

Patents (€1,750,000 - €1,600,000)	150,000
Recovery of Impairment Loss	150,000

IMPAIRMENT OF INTANGIBLE ASSETS

Impairment of Indefinite-Life Intangibles Other than Goodwill

- ◆ Should be tested for impairment at least annually.
- ◆ Impairment test is the same as that for limited-life intangibles. That is,
 - ▶ compare the recoverable amount of the intangible asset with the asset's carrying value.
 - ▶ If the recoverable amount is less than the carrying amount, the company recognizes an impairment.

IMPAIRMENT OF INTANGIBLE ASSETS

Illustration: Arcon Radio purchased a broadcast license for €2,000,000. The license is renewable every 10 years. Arcon Radio has renewed the license with the GCC twice, at a minimal cost. Because it expects cash flows to last indefinitely, Arcon reports the license as an indefinite-life intangible asset. Recently, the GCC decided to auction these licenses to the highest bidder instead of renewing them. Based on recent auctions of similar licenses, Arcon Radio estimates the fair value less costs to sell (the recoverable amount) of its license to be €1,500,000.

ILLUSTRATION 12-9
Computation of Loss on
Impairment of Broadcast License

Carrying value of broadcast license	€2,000,000
Recoverable amount (based on fair value less costs to sell)	<u>(1,500,000)</u>
Loss on impairment	€ 500,000

IMPAIRMENT OF INTANGIBLE ASSETS

Impairment of Goodwill

- ◆ Companies must test goodwill at least **annually**.
- ◆ Impairment test is conducted based on the **cash-generating unit** to which the goodwill is assigned.
 - ▶ **Cash-generating unit** = smallest identifiable group of assets that generate cash flow.
- ◆ Because there is rarely a market for cash-generating units, estimation of the recoverable amount for goodwill impairments is usually based on **value-in-use estimates**.

IMPAIRMENT OF INTANGIBLE ASSETS

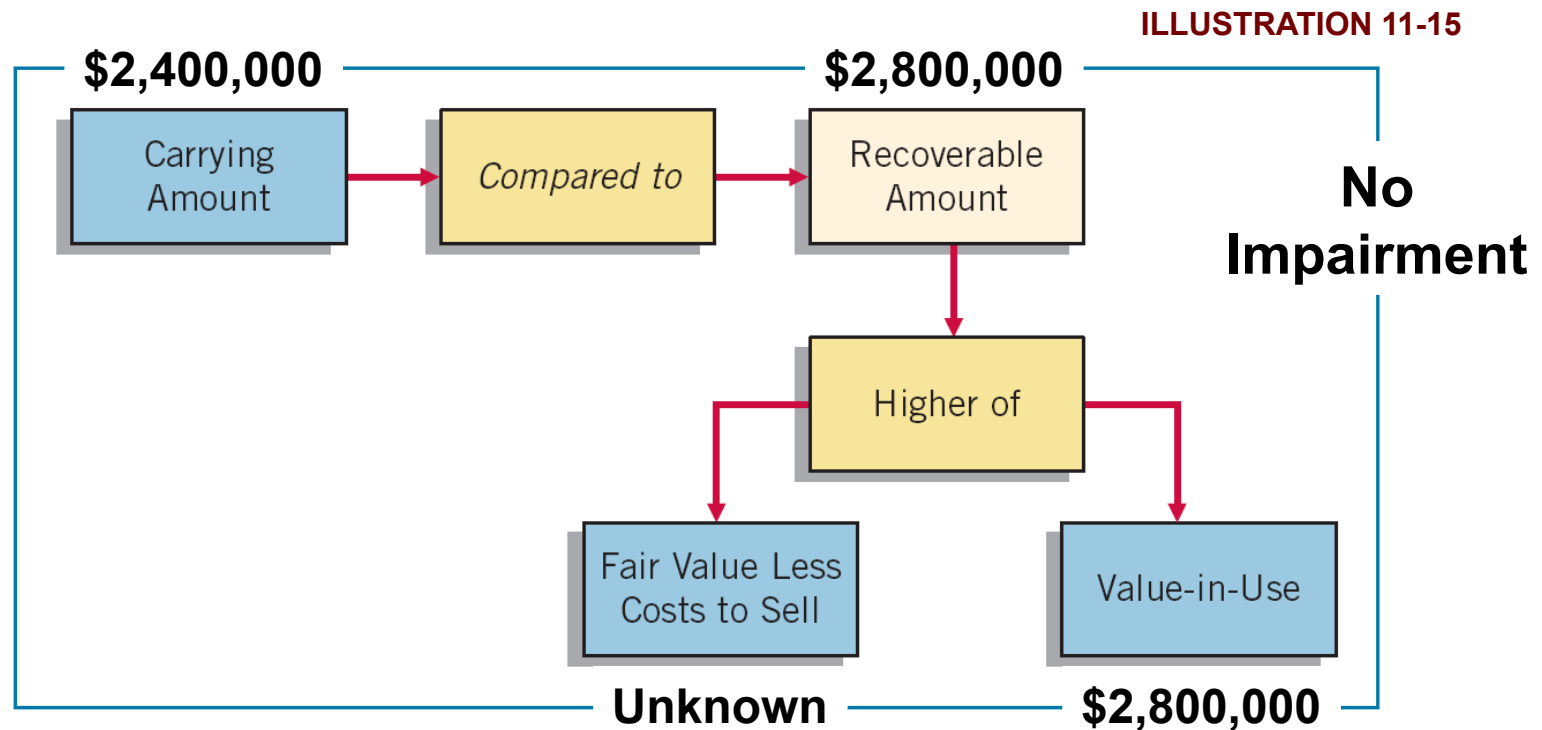
Illustration: Kohlbuy Corporation has three divisions. It purchased one division, Pritt Products, four years ago for €2 million. Unfortunately, Pritt experienced operating losses over the last three quarters. Kohlbuy management is now reviewing the division (the cash-generating unit), for purposes of its annual impairment testing. Illustration 12-10 lists the Pritt Division's net assets, including the associated goodwill of €900,000 from the purchase.

ILLUSTRATION 12-10

Property, plant, and equipment (net)	€ 800,000
Goodwill	900,000
Inventory	700,000
Receivables	300,000
Cash	200,000
Accounts and notes payable	(500,000)
Net assets	<u>€2,400,000</u>

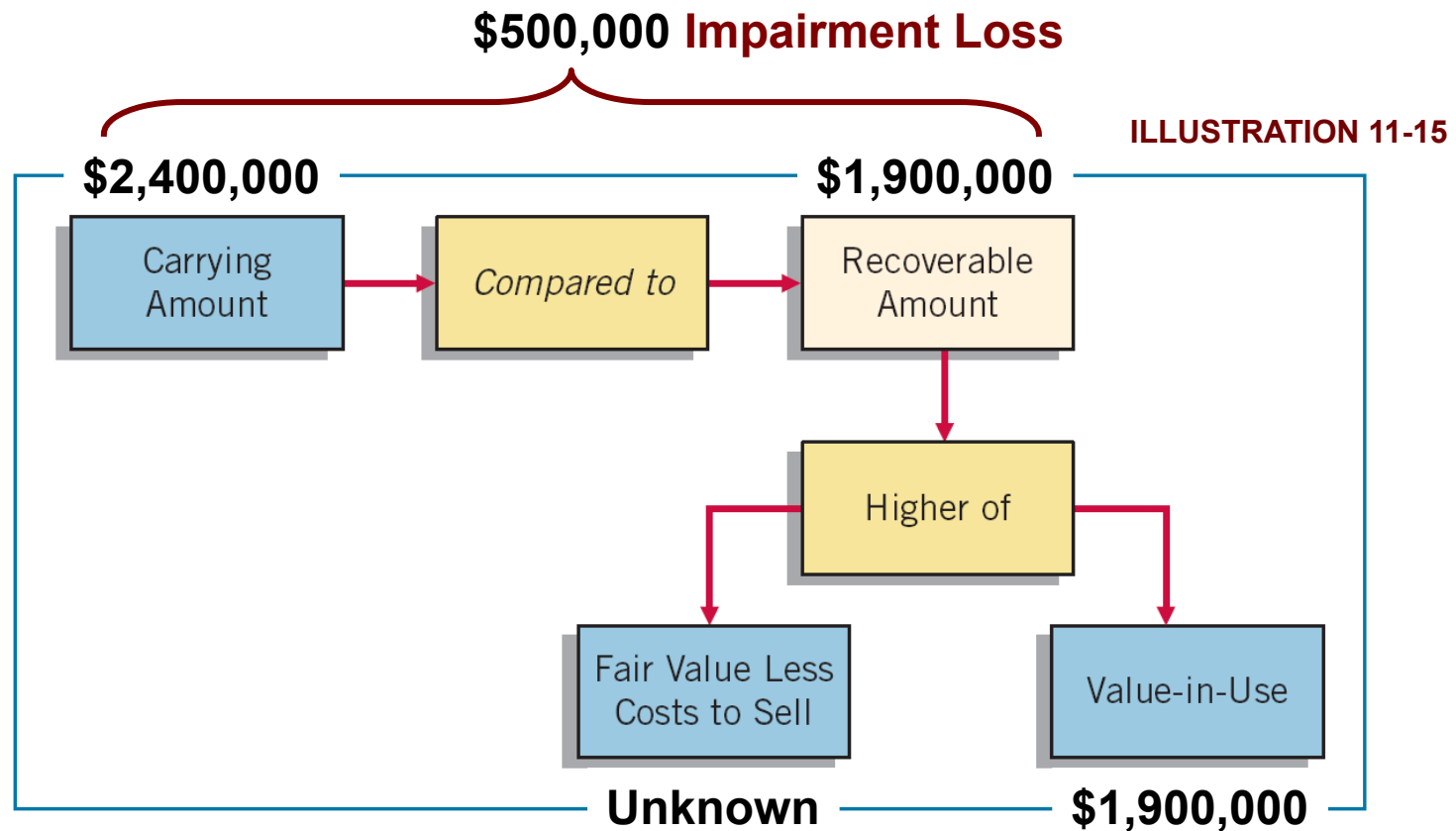
IMPAIRMENT OF INTANGIBLE ASSETS

Kohlbuy determines the recoverable amount for the Pritt Division to be €2,800,000, based on a value-in-use estimate.



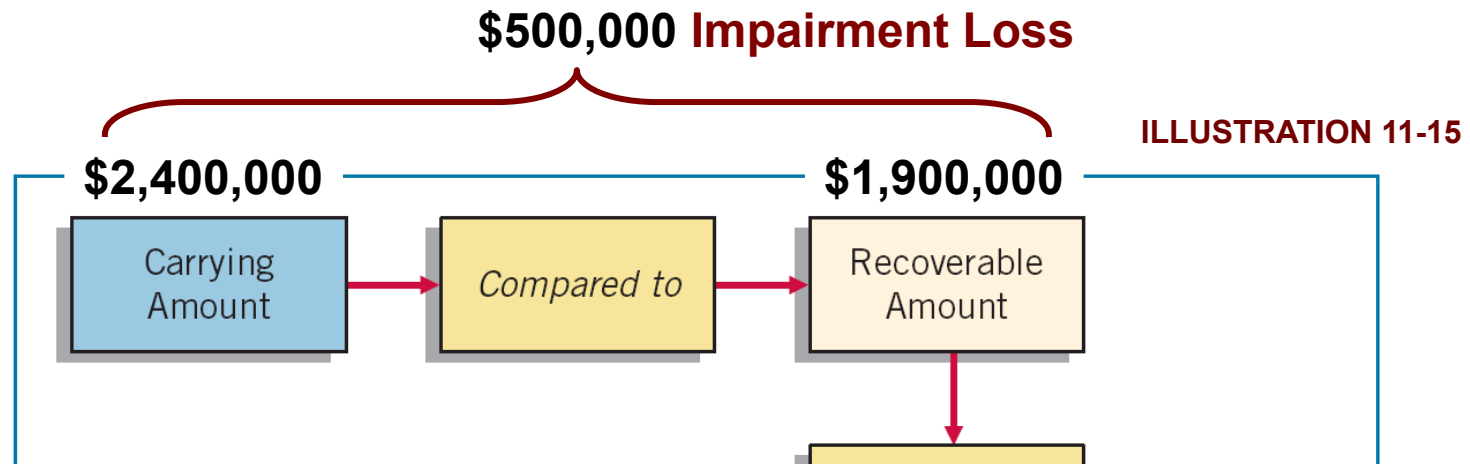
IMPAIRMENT OF INTANGIBLE ASSETS

Assume that the recoverable amount for the Pritt Division is €1,900,000 instead of €2,800,000.



IMPAIRMENT OF INTANGIBLE ASSETS

Assume that the recoverable amount for the Pritt Division is €1,900,000 instead of €2,800,000.



Kohlbuy makes the following entry to record the impairment.

Loss on Impairment	500,000
Goodwill	500,000

RESEARCH AND DEVELOPMENT COSTS

Research and development (R&D) costs are not in themselves intangible assets.

Frequently results in the development of **patents** or **copyrights** such as new

- ◆ product,
- ◆ process,
- ◆ idea,
- ◆ formula,
- ◆ composition, or
- ◆ literary work.

RESEARCH AND DEVELOPMENT COSTS

Companies spend considerable sums on **research and development**.

ILLUSTRATION 12-12

R&D Outlays, as a
Percentage of Sales

<u>Company</u>	<u>Sales (millions)</u>	<u>R&D/Sales</u>
Canon (JPN)	¥3,479,788	8.52%
Daimler (DEU)	€114,297	3.66%
GlaxoSmithKline (GBR)	£26,431	15.01%
Johnson & Johnson (USA)	\$67,224	13.13%
Nokia (FIN)	€30,176	15.85%
Roche (CHE)	CHF45,449	21.02%
Procter & Gamble (USA)	\$83,680	2.42%
Samsung (KOR)	₩201,103,613	5.73%

RESEARCH AND DEVELOPMENT COSTS

- ◆ **Research** costs must be expensed as incurred.
- ◆ **Development** costs may or may not be expensed as incurred.
- ◆ Capitalization begins when the project is far enough along in the process such that the economic benefits of the R&D project will flow to the company (the project is **economically viable**).

RESEARCH AND DEVELOPMENT COSTS

Identifying R & D Activities

ILLUSTRATION 12-13
Research Activities versus
Development Activities

Research Activities Original and planned investigation undertaken with the prospect of gaining new scientific or technical knowledge and understanding.	 Examples Laboratory research aimed at discovery of new knowledge; searching for applications of new research findings.
Development Activities Application of research findings or other knowledge to a plan or design for the production of new or substantially improved materials, devices, products, processes, systems, or services before the start of commercial production or use.	 Examples Conceptual formulation and design of possible product or process alternatives; construction of prototypes and operation of pilot plants.

RESEARCH AND DEVELOPMENT COSTS

Accounting for R & D Activities

Costs Associated with R&D Activities:

- ◆ Materials, equipment, and facilities.
- ◆ Personnel.
- ◆ Purchased intangibles.
- ◆ Contract Services.
- ◆ Indirect Costs.

RESEARCH AND DEVELOPMENT COSTS

E12-1: Indicate how items on the list below would generally be reported in the financial statements.

Item	Classification
1. Investment in a subsidiary company.	1. Long-term investments
2. Timberland.	2. PP&E
3. Cost of engineering activity required to advance the design of a product to the manufacturing stage.	2. R&D expense
4. Lease prepayment (6 months' rent).	2. Prepaid rent
5. Cost of equipment obtained.	2. PP&E
6. Cost of searching for applications of new research findings.	2. R&D expense

RESEARCH AND DEVELOPMENT COSTS

Item	Classification
7. Cost incurred in the formation of a corporation.	7. Expense
7. Operating losses incurred in the start-up of a business.	7. Operating loss
7. Training costs incurred in start-up of new operation.	7. Expense
7. Purchase cost of a franchise.	7. Intangible
7. Goodwill generated internally.	7. Not recorded
7. Cost of testing in search of product alternatives.	7. R&D expense

RESEARCH AND DEVELOPMENT COSTS

Item	Classification
13. Goodwill acquired in the purchase of a business.	13. Intangible
13. Cost of developing a patent (before achieving economic viability).	13. R&D expense
13. Cost of purchasing a patent from an inventor.	13. Intangible
13. Legal costs incurred in securing a patent.	13. Intangible
13. Unrecovered costs of a successful legal suit to protect the patent.	13. Intangible

RESEARCH AND DEVELOPMENT COSTS

Item	Classification
18. Cost of conceptual formulation of possible product alternatives.	18. R&D expense
18. Cost of purchasing a copyright.	18. Intangible
18. Development costs incurred after achieving economic viability.	18. Intangible
18. Long-term receivables.	18. Long-term investment
18. Cost of developing a trademark.	18. Expense
18. Cost of purchasing a trademark.	18. Intangible

RESEARCH AND DEVELOPMENT COSTS

Costs Similar to R & D Costs

- ◆ Start-up costs for a new operation.
- ◆ Initial operating losses.
- ◆ Advertising costs.

These costs are expensed as incurred, similar to the accounting for R&D costs.

RESEARCH AND DEVELOPMENT COSTS

E12-17: Compute the amount to be reported as **research and development expense**.

$$\$330,000 / 5 = \$66,000$$

R&D
Expense

Cost of equipment acquired that will have alternative uses in future R&D projects over the next 5 years.

\$330,000 \$66,000

Materials consumed in R&D projects

59,000 59,000

Consulting fees paid to outsiders for R&D projects

100,000 100,000

Personnel costs of persons involved in R&D projects

128,000 128,000

Indirect costs reasonably allocable to R&D projects

50,000 50,000

Materials purchased for future R&D projects

34,000 0

\$403,000

PRESENTATION OF INTANGIBLES

Presentation of Intangible Assets

Statement of Financial Position

- ◆ Companies should report as a separate item all intangible assets other than goodwill.
- ◆ Reporting is similar to the reporting of property, plant, and equipment.
- ◆ Contra accounts may not normally shown for intangibles.

PRESENTATION OF INTANGIBLES

Presentation of Intangible Assets

Income Statement

Companies should report

- ◆ amortization expense and
- ◆ impairment losses and reversals

for intangible assets other than goodwill separately in net income (usually in the operating section).

Notes to the financial statements should include the amortization expense for each type of asset.

PRESENTATION OF INTANGIBLES

Presentation of Intangible Assets

ILLUSTRATION 12-15
Nestlé's Intangible Asset
Disclosures



Nestlé

(CHF 000,000)

INCOME STATEMENT
FOR THE YEAR ENDED 31 DECEMBER 2012

Amortization expense	439
Impairments of goodwill	14

BALANCE SHEET
AT 31 DECEMBER 2012

Goodwill (note 14)	32,615
Intangible assets	13,643

PRESENTATION OF INTANGIBLES

Presentation of Research and Development Costs

Companies should disclose the total R&D costs charged to expense each period.

ILLUSTRATION 12-16
R&D Reporting



GlaxoSmithKline
(000,000)

31 December 2012

Turnover	£26,431
Cost of Sales	(7,894)
Gross Profit	18,537
Selling, General, and Administrative	(8,739)
Research and Development	(3,968)
Other Operating Income	1,562
Operating Profit	£ 7,392